

Yissum Research Intelligence Report — Week 30, 2026

7 high-potential paper(s) · Weekly · Week 30, 2026 · Generated July 22, 2026

This digest highlights pioneering advances across several high-impact fields, offering significant commercial potential for investors and industry partners. Featured innovations include novel cell and immunotherapies tackling neurological disorders and cancer, alongside transformative platform technologies in genomics, drug delivery, and AI-driven bioinformatics. Additionally, a scalable solution for enhanced hospital infection control underscores a strong market for immediate application.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Tamir Ben-Hur** — Neuroscience, Immunology, Clinical, Synthetic Biology
- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Oren Zimhony** — Medical Device, Materials, Clinical
- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Mor Nitzan** — Drug Discovery, Software/AI, Genomics, Immunology
- **Ofer Mandelboim** — Immunology, Drug Discovery, Clinical
- **Ayelet N Landau** — Diagnostics, Medical Device, Software/AI, Neuroscience

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders. 36/50

"Stem cell-derived astrocyte therapy shows promise for repairing myelin in neurological disorders."

Main researcher: Tamir Ben-Hur · tamir@hadassah.org.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; The Department of Neurology, The Agnes-Ginges Center for Human Neurogenetics, Hadassah-Hebrew University Medical Center, Jerusalem, Israel. Electronic address: tamir@hadassah.org.il.

Published in Stem cell reports · 2026-06

ABOUT THIS RESEARCH

Researchers developed a clinical-grade astrocyte therapy derived from embryonic stem cells that helps the brain repair damaged myelin sheaths. The therapy works by clearing debris and creating a supportive environment that encourages the growth of new insulating layers for nerves.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

COMMERCIALISATION METRICS

Novelty

6/10

While the use of hESC-derived astrocytes is a sophisticated approach to remyelination, the 'bystander mechanism' and the focus on microglial debris clearance are established concepts in neuroregeneration, making this more of an incremental application of cell therapy than a groundbreaking scientific paradigm shift.

Commercial Potential

8/10

The therapy uses clinical-grade, scalable hESC-derived cells targeting a high-unmet-need market (demyelinating disorders) with demonstrated safety and a clear mechanism of action. The 'clinical-grade' status and scalability suggest it is already positioned for translational development rather than being purely theoretical.

Market Size

9/10

The therapy targets demyelinating disorders (such as Multiple Sclerosis), which represent a massive, high-value patient population with significant unmet medical needs. The mechanism's ability to address multiple facets of remyelination failure suggests a broad, multi-indication platform potential within the neurodegenerative market.

Tech Readiness

5/10

The technology has transitioned from basic lab discovery to 'clinical-grade' manufacturing and has demonstrated efficacy in animal models (lysolecithin-induced), but it remains in the preclinical validation stage rather than undergoing human clinical trials.

IP Strength

8/10

The development of a 'clinical-grade' hESC-derived astrocyte product (AstroRx) suggests a proprietary, scalable manufacturing process and a specific cell composition that is highly patentable. The multi-modal mechanism of action—addressing debris clearance and OPC differentiation—provides multiple layers of therapeutic defensibility and platform potential for various CNS disorders.

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2. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method **35/50** for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells.

"Next-gen single-cell sequencing reveals hidden disease mechanisms, boosting precision diagnostics."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

ABOUT THIS RESEARCH

Researchers have developed a new single-cell sequencing technology called sc-rDSeq that captures a much wider range of RNA types, including non-polyadenylated RNAs, at a lower cost and higher density than current standards. By applying this to lung cancer, they demonstrated the ability to identify hidden cell subpopulations and drug-resistance mechanisms that traditional methods miss.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty

8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential

8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size

8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness

4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength

7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

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3. The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms. 33/50

"Novel coating extends disinfectant power, sharply cutting hospital-acquired infections."

Main researcher: Oren Zimhony · Oren_Z@clalit.org.il.

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Published in Antimicrobial resistance and infection control · 2026-06

ABOUT THIS RESEARCH

Researchers tested OxiLast™, a specialized surface coating that extends the antimicrobial effectiveness of chlorine-based disinfectants in hospital isolation rooms. The study demonstrated that applying this coating alongside standard cleaning significantly reduces the presence of multidrug-resistant bacteria like CRAB and MRSA on high-touch surfaces.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

COMMERCIALISATION METRICS

Novelty

5/10

The research presents an incremental improvement on existing disinfection protocols by using a proprietary coating to extend the efficacy of standard chlorine-based agents. While clinically useful, the innovation is an additive formulation rather than a groundbreaking paradigm shift in antimicrobial science.

Commercial Potential

8/10

The technology demonstrates a clear, additive value proposition to existing hospital disinfection protocols and uses a validated proof-of-concept to address high-value clinical needs (MDRO reduction). The 'proprietary formulation' suggests protectable IP that can be easily commercialized as a specialized chemical adjunct for the healthcare sector.

Market Size

6/10

The technology addresses a high-value niche in infection control (MDRO isolation rooms), but its application as an adjunct to existing protocols suggests a specialized additive market rather than a broad-scale replacement for global disinfection commodities.

Tech Readiness

7/10

The technology has moved beyond lab-scale validation into a controlled quasi-experimental clinical setting with real-world implementation in hospital isolation rooms. It is currently at the stage of clinical validation/pilot testing, positioned just before full-scale commercial rollout or large-scale multi-center trials.

IP Strength

7/10

The use of a proprietary formulation (OxiLast™) to stabilize existing disinfectants suggests a protectable chemical composition or method-of-use patent. However, the efficacy relies on an additive application to standard protocols, which may face moderate defensibility challenges against generic formulations or easier-to-replicate chemical stabilizers.

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4. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake.

33/50

"Revolutionary synthetic DNA analogue enters cells naturally, enabling advanced gene therapies."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

ABOUT THIS RESEARCH

Researchers developed a new type of synthetic DNA analogue called MCP PNA that can naturally enter cells without needing complex delivery vehicles. This solves a major historical bottleneck where these stable molecules were too large or poorly charged to cross cell membranes.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty

6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential

8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size

9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness

3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength

7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

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5. Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck. 28/50

"AI-powered software deciphers complex single-cell data, accelerating drug discovery insights."

Main researcher: Mor Nitzan · mor.nitzan@mail.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel; Racah Institute of Physics, The Hebrew University of Jerusalem, Jerusalem, Israel; Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.

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Published in Cell systems · 2026-05

ABOUT THIS RESEARCH

This paper introduces bioIB, an AI framework that uses the information bottleneck method to distill complex single-cell RNA sequencing data into highly specific, interpretable gene clusters. By filtering out noise and focusing on relevant biological signals like disease states, it maps out the hierarchical relationships between different cell types and biological processes.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

COMMERCIALISATION METRICS

Novelty

6/10

The paper applies an established information theory framework (Information Bottleneck) to a well-trodden domain (scRNA-seq), representing an incremental methodological improvement rather than a fundamental scientific breakthrough. While the application to gene clustering and hierarchical structure is useful, it follows existing trends in deep learning-based dimensionality reduction and feature selection.

Commercial Potential

6/10

The framework presents a strong platform-enabling computational tool for drug discovery and biomarker identification, but its commercial value is currently tied to software/algorithm licensing rather than a standalone proprietary product. Its utility depends on integration into existing bioinformatics pipelines or single-cell analysis platforms.

Market Size

8/10

The framework acts as a platform-enabling technology for the massive and growing single-cell multi-omics market, offering high value for drug discovery and precision medicine. By optimizing signal extraction in complex datasets, it addresses a critical bottleneck in the multi-billion dollar bioinformatics and clinical diagnostics sectors.

Tech Readiness

4/10

The paper presents a computational framework and algorithmic methodology (bioIB) rather than a physical product or clinical tool. While it demonstrates utility on existing datasets, it remains at the fundamental software/algorithm development stage (TRL 3-4) rather than a validated industrial platform.

IP Strength

4/10

The innovation is primarily a computational framework based on the Information Bottleneck method, which is an established theoretical principle in machine learning, making the core algorithm difficult to patent. While the application to scRNA-seq is valuable, the defensibility is low as it likely falls under 'mathematical methods' and is susceptible to rapid replication by open-source community developments.

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6. RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

"Bacterial protein boosts immune cells to fight cancer, a new immunotherapy avenue."

Main researcher: Ofer Mandelboim

The Concern Foundation Laboratories at the Lautenberg Center for Immunology and Cancer Research, Institute for Medical Research Israel Canada (IMRIC), Hebrew University Hadassah Medical School, Jerusalem, Israel.

Published · 2026-05

ABOUT THIS RESEARCH

Researchers discovered that a specific protein (RadD) from the bacterium *F. nucleatum* can activate NKp46 receptors on natural killer cells to kill tumor cells. This reveals a complex relationship where certain bacterial components can actually trigger the immune system to fight cancer.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

COMMERCIALISATION METRICS

Novelty

7/10

The discovery of a specific bacterial protein (RadD) acting as a direct ligand for the NKp46 receptor is highly novel and expands the known ligandome of NK cells. While it provides a strong foundation for targeted immunotherapies, it is an incremental discovery of a molecular interaction rather than a platform-level technological breakthrough.

Commercial Potential

6/10

The identification of RadD as a specific ligand for NKp46 provides a clear molecular target for developing NK cell agonists or mimics to enhance antitumor cytotoxicity. However, the commercial path is complicated by the dual nature of *F. nucleatum* in cancer progression and the need for precise delivery to avoid systemic inflammation.

Market Size

6/10

The target market encompasses oncology immunotherapy, specifically head and neck cancers, which is substantial; however, the current focus is on a highly specific host-microbe interaction rather than a broad platform application.

Tech Readiness

3/10

The research is at the fundamental discovery stage, establishing a biological mechanism through in vitro assays and murine models. While it identifies a novel target for immunotherapy, there are no lead candidates, pharmacokinetic data, or clinical validation steps presented.

IP Strength

6/10

The identification of a specific bacterial ligand (RadD) for a known human receptor (NKp46) provides a clear target for composition-of-matter patents on RadD-mimetics or agonists. However, since RadD is a naturally occurring bacterial protein, the IP strength depends on the ability to engineer novel derivatives or specific delivery vehicles rather than the discovery of the axis itself.

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7. Universal rhythmic architecture uncovers two modes of neural dynamics.28/50

"Universal brain activity patterns enable personalized neurodiagnostics and BCI."

Main researcher: Ayelet N Landau

Department of Psychology, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Cognitive and Brain Sciences, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Experimental Psychology, University College London (UCL), London, WC1H 0AP, UK.
Published in Nature communications · 2026-05

ABOUT THIS RESEARCH

Researchers have discovered a universal pattern in brain activity that distinguishes between sustained rhythms and brief, transient bursts. This framework allows for a more nuanced understanding of how different brain regions process ongoing information versus responding to new environmental stimuli.

COMMERCIAL ANGLE

This universal architecture can be used to develop highly personalized neurodiagnostic software and advanced brain-computer interfaces that distinguish between stable cognitive states and rapid sensory responses.

COMMERCIALISATION METRICS

Novelty8/10

The research shifts the fundamental paradigm of neural analysis from a frequency-only model to a two-dimensional architecture (frequency + rhythmicity), providing a potentially transformative framework for neurotechnological diagnostics. This provides a platform-enabling discovery that unifies disparate datasets across species and clinical states into a single organizing principle.

Commercial Potential5/10

While the framework offers high potential for improving the precision of neurodiagnostic biomarkers and personalized EEG analysis, it remains a fundamental descriptive discovery rather than a specific device or therapeutic intervention. Its commercial value depends entirely on its integration into future clinical software or neurotechnology platforms.

Market Size8/10

The findings provide a platform-enabling framework for universal neurophysiological analysis, addressing a massive market spanning clinical diagnostics, neuroprosthetics, and brain-computer interfaces across diverse patient populations.

Tech Readiness3/10

The paper presents fundamental discovery-based neuroscience that provides a new theoretical framework for understanding neural dynamics. While it offers potential for future diagnostic biomarkers, it is currently at a basic research stage without a defined prototype, hardware integration, or clinical validation pipeline.

IP Strength4/10

The work describes a scientific framework and a descriptive method for analyzing existing neural data rather than a novel hardware device or a proprietary biochemical entity. While the 'rhythmicity measure' could potentially be patented as a diagnostic algorithm, the primary output is foundational knowledge that is difficult to defend against competitors using alternative mathematical modeling.

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